



Department of Telecommunications
Telecommunication Engineering Centre
110001 –K.L. Bhawan, Janpath, New Delhi
(Convergence & Broadcasting Division)



No: - 4-16/2018-C&B/TEC/NWG-9

dated: 15.07.2022

Sub: Minutes of Meeting held on 13-07-2022 of the National Working Group 9 (NWG-9) corresponding to ITU-T Study Group 9 (SG9) for the Study Period 2022- 2024.

The third meeting of the reconstituted National Working Group-9 (NWG-9) corresponding to the ITU-T Study Group 9 (SG9) for Study Period 2022-24 regarding "Audio visual content transmission and integrated broadband cable networks" or "Broadband Cable and TV" was held on 13-07-2022 under the chairmanship of Sh. Avinash Agarwal, DDG (Convergence & Broadcasting), TEC through video-conferencing. List of participants is placed at Annexure-I.

2. At the outset, Sh. Rakesh Desai, Director (C&B-II), TEC & Convenor, NWG9 welcomed all the participants and briefly explained them about the objectives and scope of NWG-9. He explained the members about the Agenda of the meeting. He also requested all the participants to submit their contributions by 25th July, 2022.
3. Shri Avinash Agarwal, Chairman NWG-9 & DDG (C&B), TEC, in his introductory remarks, explained the discussions of the previous two meetings of NWG-9. He requested all the participants to introduce themselves keeping in view that some of the participants were participating for the first time in the meeting. Accordingly, all the participants gave their brief introduction and shared information about their interest areas of Work.
4. After the introduction Sh. Avinash Agarwal presented a brief introduction and purpose of the SG9 and NWG-9. He discussed about Indian contributions for the SG9 meeting scheduled from 6th to 14th September 2022. He shared the screen of ITU-T Study Group 9 website and discussed about the mandate of the SG9, questions under study and current work items. Based on the discussions in the previous two meetings and subsequent interaction with the members, he listed following areas wherein NGW-9 plans to contribute to SG-9:-
 - i. Recommendation of Television broadband direct to mobile (D2M).
 - ii. Recommendation of Television distribution using public Wi-Fi.
 - iii. Methods and practices for conditional access and content protection for cable television.
 - iv. AI-enabled enhanced functions over integrated broadband cable network.
 - v. Common user profile format for audio-visual content distribution.

Sh. Avinash Agarwal also clarified that members may furnish their contributions on any other questions and work items of SG-9 over and above the aforesaid identified

topics and may also add new work items relevant to the listed questions of SG-9 as per their interests and expertise.

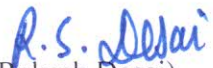
5. Dr. Pradipta Biswas, IISc Bengaluru & Vice-Chair NWG-9, said that he is working on Q. No. -11/9 of SG-9 on accessibility and shall be happy to share the information to the interested members who want to contribute on it. He also clarified on the difference of common user profile format of SG-9 vis-à-vis SG-16 and said that contribution on this question will be coming through SG-9 only although we are being collaborating with SG-6 & 16 in this matter. It will be a joint work of SG-6, SG-9 and SG-16. He also assured to apprise NWG-9 members about discussions in SG-9 meetings on various questions.
6. Ms. Meenaksi said the they had a conclave recently on D2M with IIT-K, Prasar Bharti and TSDSI wherein proof of concept (PoC) was demonstrated as to how TV signal can be broadcast directly to mobile phones utilising spectrum of free band (due to scrapping of Analogue terrestrial transmission) more effectively.
7. Sh. Manav, representative of Sh.Pallab Dutta, C-DoT said that they are contributing on television distribution though public Wi-Fi and have submitted the initial version document to the Chairman NWG-9. He also briefly explained about their contribution in this regard during the meeting.
8. Sh. R. Venketesh said that they are working on Q.no. 1/9 of ITU-T regarding secondary distribution of signal and shall submit the contribution if found appropriate.
9. Sh.Rajiv Khattar said that they are working on Q. no.11/ 9 on accessibility to disabled and shall submit the contribution in due course.
10. Sh. Sultan Singh from M/s Samsung informed that they could not contribute directly. However, he assured that he will look into as to how M/s Samsung can contribute in this group.
11. Sh. Sunder Naryanan said that he is working on Q. No-3/9 on AI enabled enhanced functions over integrated Broadband cable network, responsible use of technology & data handling and validation methods. He also explained about fair use of AI, use of behaviour biometric data, preferences and choices made on the basis of inferences, etc to the participants.
12. Prof Rajendra Kumar Pandey, IIT BHU, informed that he will submit his inputs separately.
13. Sh. Avinash Agarwal highlighted that TEC has recently published test guidelines for Conditional Access System (CAS) and Subscriber Management System (SMS) on its website and the interested members may please go through it as it has relevance to the contribution regarding methods and practices for conditional access and content protection for cable television (Q.no.2/9).

14. Sh. Avinash Agarwal apprised about various timelines of furnishing contributions to SG-9. He requested to all the participants to send their contribution by the 25th July 2022 for further deliberations in NWG-9 group.
15. Sh. Avinash Agarwal also requested to the members to identify relevant experts from the Academia, start-ups, industry, OEMs, etc by next week so that such experts can be included in NWG-9 group.
16. Sh. Avinash Agarwal summarised the interests shown by members in submitting contributions saying that Ms. Meenaksi will contribute related to D2M; C-DOT will contribute related to public wi-fi; Sh. Sunder Naryanan will work on contribution related to AI-enabled enhanced functions; Dr. Pradipta Biswas in association with Sh. Rajiv Khattar will work on the contribution related to Common user profile format and accessibility; Sh. R. Venketesh may work on contribution related to Q.No.1/9 regarding secondary distribution of signal and TEC members will work on contributions related to conditional access & content protection. He also encouraged other members to give their inputs on the above-stated areas to the members who are working on it. He further suggested to create focussed working sub-groups within NWG9 so that quality contributions can be prepared in a time-bound manner.

17. The key action points emerged from the meeting are as under:-

- a. NWG-9 members may furnish their contributions on the questions/ work items pertaining to SG-9 as per their interest/ commitment by **25th of July 2022.**
- b. NWG-9 members may suggest names of experts from the Academia, start-ups, industry, OEMs that can be added in the group.
- c. The members may act as facilitators and encourage relevant experts/ researchers to contribute.

18. The meeting ended with the vote of thanks to the chair.


(Rakesh Desai)
Convener, NWG-9 &
Director, (C&B-II), TEC

Encl.: Annexure-I

To,
All members of the NWG-9

Copy for kind information to:
Sr.DDG & Head TEC, New Delhi

ANNEXURE-I

List of Participants:

1. Sh. Avinash Agarwal, Chairman NWG-9 and DDG C&B TEC.
2. Dr. Pradipta Biswas, IISc Bengaluru & Vice-Chair NWG-9.
3. Prof. Rajendra Kumar Pandey, IIT BHU.
4. Ms. Meenakshi, Director (Engg.), AIR & Doordarshn (WR).
5. Major General Dr. Pritam Bisnoi, TSDSI.
6. Sh. Manav, C-DoT.
7. Sh. Rajiv Khattar, , Researcher & Industry specialist
8. Sh. Subhash Kumar, DS, MIB.
9. Sh. Sultan Singh, M/s Samsung.
10. Sh. Navin Kumar, M/s Samsung.
11. Sh. Sunder Narayan, Researcher/ Consultant
12. Mr. R. Venkatesh, Director (Engineering), Doordarshan
13. Sh. Sunil Gautam, Director (Social Media), D/o Telecom.
14. Sh. Venkata Rama Raju Chelle, Director (6GT) TEC
15. Sh. S. K. Arya, Director (MRA), TEC
16. Sh. Naveen Khichar, ADET, DOT.
17. Sh. Brajesh Kumar, Director (C&B-I) TEC.
18. Sh. Rakesh Desai, Convener, NWG-9 & Director (C&B-II)
19. Sh. Balwant Rai, ADG (C&B), TEC.
20. Ms. Jyoti Prajapati, Intern, TEC.